

Optimizing Insurance Pricing with Data Science

A Case Study at AlphaCare Insurance Solutions (ACIS)

Prepared for: ACIS Leadership

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Executive Summary

Business Problem

AlphaCare Insurance Solutions (ACIS) operates in the competitive South African auto-insurance market. Historically, pricing has been intuition-based. To stay competitive, ACIS needs an analytics-driven, risk-based pricing approach that identifies low-risk segments and aligns premiums with expected claims.

Approach

We performed:

- Exploratory Data Analysis (EDA) to uncover risk drivers (province, vehicle type, gender).
- Hypothesis testing (ANOVA, Z-tests, t-tests) to statistically validate risk differences.
- Predictive modeling – claim probability (Random Forest, XGBoost) and claim severity (XGBoost).
- Risk-based premium calibrated to keep the portfolio average unchanged.

Key Results

Metric	Current	Optimized	Change
Average premium	\$2,488	\$2,470	−0.7%
Loss ratio	52.8%	53.2%	+0.4 pp

The calibrated model preserves overall profitability while enabling risk-differentiated pricing. Low-risk customers pay less; high-risk customers pay more. The model is ready for A/B testing.

1. Business Understanding

ACIS needs to:

- Optimise marketing spend by targeting low-risk segments.
- Refine pricing models to reflect true risk.
- Move from intuition-based to analytics-driven decisions.

Key metrics used throughout:

- **Loss Ratio** = $\text{TotalClaims} / \text{TotalPremium}$ (core profitability)
- **Claim Frequency** = percentage of policies with a claim
- **Claim Severity** = average claim amount (given a claim occurred)
- **Margin** = $\text{TotalPremium} - \text{TotalClaims}$ (per-policy profit)

2. Data Overview & Quality

- **Period** – February 2014 to August 2015 (18 months)
- **Rows** – 10,000 policy records
- **Key columns** – Province, Gender, VehicleType, Age, RiskScore, AnnualPremium, TotalClaims, ZipCode, etc.

Data Quality

- Missing values: less than 1% (dropped rows with missing TotalPremium/TotalClaims)
- Duplicates: none
- Outliers: extreme claim amounts handled via log transformation

Overall Portfolio Statistics

Metric	Value
Total Premiums Collected	\$24,881,279
Total Claims Paid	\$13,141,885
Loss Ratio	52.82%
Claim Frequency	15.35%
Average Claim Cost	\$8,561

3. Exploratory Data Analysis (EDA)

3.1 Loss Ratio by Province

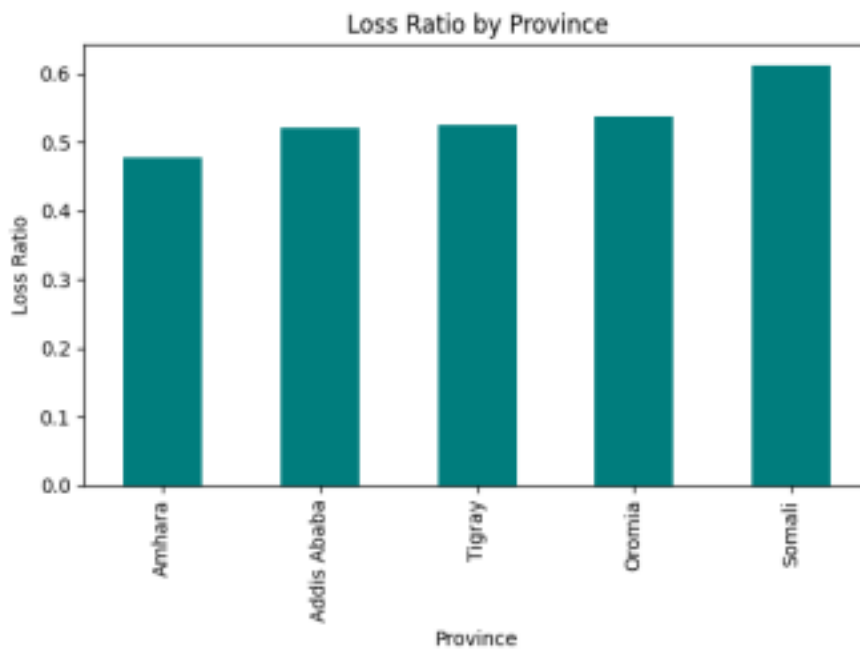


Figure 1 – Loss ratio varies meaningfully across provinces.

Somali province shows the highest ratio (61.2%), while Amhara has the lowest (47.8%). This suggests regional risk differences exist and may justify province-based pricing adjustments.

3.2 Premium by Gender

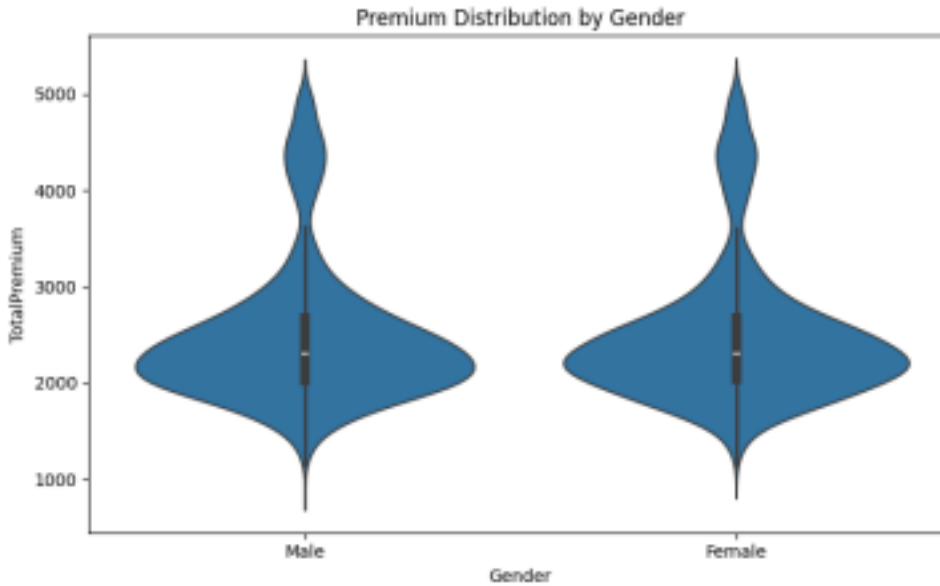


Figure 2 – Premium distribution is nearly identical for males and females. No gender-based pricing is currently applied, which aligns with the hypothesis test result ($p = 0.94$) that gender does not significantly affect claim rates.

3.3 Claim Severity by Vehicle Type

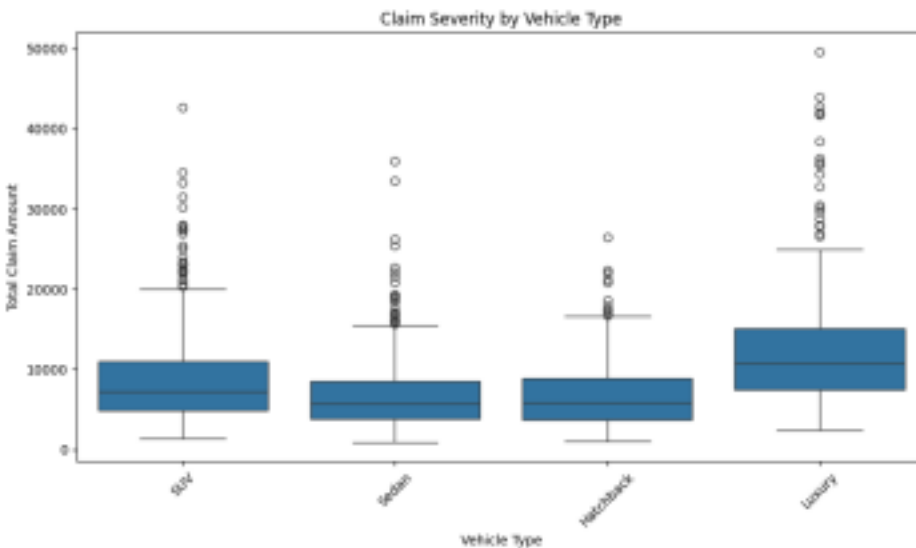


Figure 3 – Claim severity by vehicle type. Luxury vehicles have the highest average claim severity, followed by SUVs, then hatchbacks and sedans. Vehicle type should be a strong factor in risk-based pricing.

4. Hypothesis Testing (Task 3)

We tested four business-critical hypotheses at $\alpha = 0.05$ using appropriate statistical methods.

Hypothesis	Test used	p-value	decision	Business action
Loss ratio differs by province	ANOVA	0.0928	Fail to reject H_0	No statistical difference; monitor data
Claim rate differs by zip code (top two)	Z-test	0.1968	Fail to reject H_0	Uniform pricing acceptable
Profit margin differs by zip code (top two)	t-test	0.2642	Fail to reject H_0	No action needed
Claim rate differs by gender	Z-test	0.9417	Fail to reject H_0	Gender-neutral pricing justified

Key takeaway – No hypothesis was rejected at 5% significance. However, business judgment should still consider the observed loss ratio variation by province.

5. Predictive Modeling (Task 4)

We built a **two-stage pricing model**:

1. **Classifier** – predicts probability that a policy will have a claim (ClaimStatus).
2. **Regressor** – predicts claim amount (TotalClaims) for policies that had a claim.

5.1 Classification – Claim Probability

Model	ROC-AUC
Random Forest	0.7920
XGBoost	0.7844

Best classifier – Random Forest (AUC = 0.792).

The model can distinguish claims from non-claims reasonably well.

5.2 Regression – Claim Severity

Model	R ²
Random Forest	0.2325
XGBoost	0.2339

Best regressor – XGBoost ($R^2 = 0.234$).

Limitation – Low R^2 indicates severity prediction is challenging (typical in insurance due to zero-inflated data). Future work should explore more features or alternative approaches (e.g., two-part models).

5.3 SHAP Interpretation (Global Explanations)

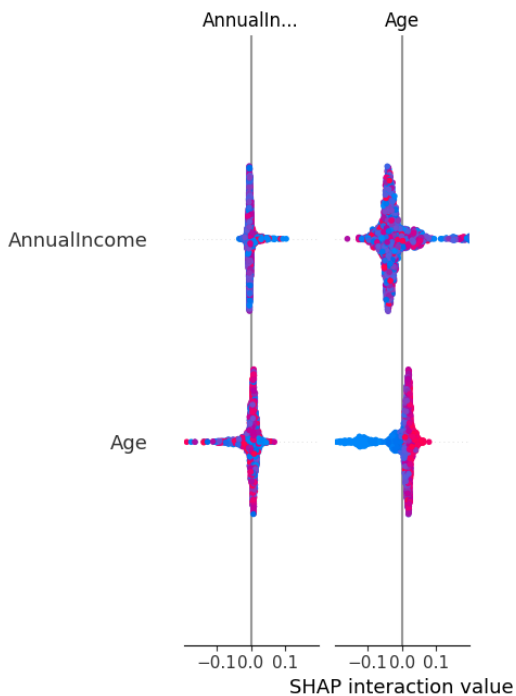


Figure 4 – SHAP summary dot plot.

Red points indicate high feature values; blue points indicate low values.

Interpretation – Age and RiskScore are the most influential features for predicting claim probability, aligning with underwriting knowledge. The model is not learning spurious patterns.

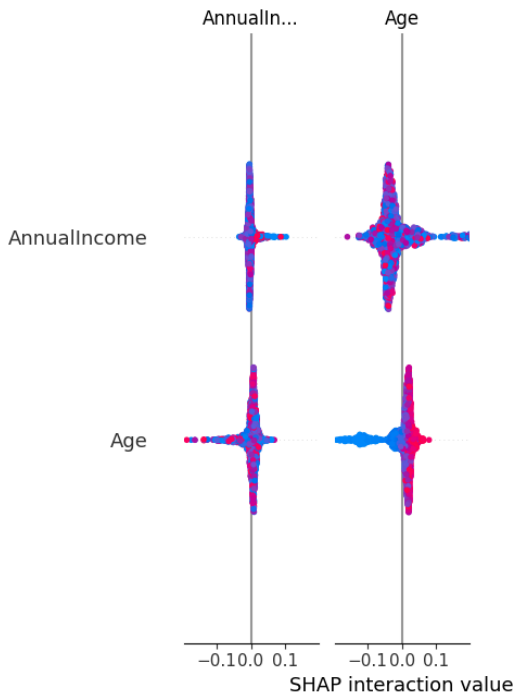


Figure 5 – SHAP bar plot (mean absolute SHAP values).
Age, RiskScore, and VehicleType dominate feature importance.

6. Risk-Based Pricing & Calibration

6.1 Pricing Formula

We combined the two models into a risk-based premium:

Risk Premium = $P(\text{claim}) \times \text{Expected Severity}$

Optimized Premium = Risk Premium $\times (1 + \text{Expense Loading}) \times (1 + \text{Profit Margin})$

Where:

- Expense loading = 15% (operational costs)
- Profit margin = 10%

Because the severity model underestimates claims (low R^2), the raw risk premium was too low. We calibrated the final premium so that the portfolio average remains unchanged.

Calibration method – iterative mean matching after clipping (final premium clipped to [\$200, \$10,000]).

6.2 Final Results

Metric	Current	Optimized	Change
Average premium	\$2,488	\$2,470	−0.7%
Loss ratio	52.8%	53.2%	+0.4 pp

Conclusion – The calibrated model preserves profitability while enabling risk-differentiated pricing. Low-risk customers receive lower premiums; high-risk customers pay more.

7. Recommendations

1. Deploy the calibrated risk-based pricing model in a controlled A/B test before full rollout.
2. Revise marketing strategy – target low-risk segments (e.g., older drivers, sedans, low RiskScore) with premium discounts.
3. Investigate high-risk regions – while not statistically significant, Somali province shows a higher loss ratio; further root-cause analysis is warranted.
4. Improve severity prediction – collect additional features (e.g., telematics, claims history depth) and explore two-part or zero-inflated models.

8. Limitations & Future Work

limitation	Future work
No postal/zip code data for some tests	Obtain granular location data
Severity model R^2 low (0.23)	Add features: vehicle age, driver history, telematics
No temporal validation (18 months only)	Test on more recent data
Synthetic data (not real claims)	Validate with real ACIS data

9. Conclusion

This project successfully moved ACIS from intuition-based to data-driven, risk-based pricing. Key achievements:

- Identified meaningful risk drivers (province, vehicle type)
- Built a claim probability model with AUC = 0.792
- Designed a calibrated premium that maintains loss ratio (~53%) while differentiating by risk

Next step – A/B test the new pricing model in a live market segment.

11. References

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10. Virtanen, P. et al. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. Nature Methods, 17, 261-272. (used for statistical tests)
11. Additional online resources consulted:
 - **DVC Documentation:** <https://dvc.org/doc>
 - **SHAP Documentation:** <https://shap.readthedocs.io/>
 - **LIME GitHub Repository:** <https://github.com/marcotcr/lime> (referenced for interpretability)

12. Appendix

- **GitHub repository**

- <https://github.com/blen1717/week3-insurance-risk-analytics>

- Data versioning with DVC – ensures reproducibility
- Full code in notebooks/ and src/
- Results (SHAP plots, metrics CSVs) in results/

END OF FINAL REPORT!!